

A ridge-regression worked example for concrete compressive-strength manuscript drafting

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Abstract

This worked example demonstrates a conservative manuscript-drafting workflow for a concrete compressive-strength prediction task. The workflow is designed around the variable schema of the UCI Concrete Compressive Strength dataset, which is a widely used public benchmark for concrete-strength modelling; however, the current local artifacts are labelled `DETERMINISTIC_FALLBACK_NOT_UCI` and must not be interpreted as verified UCI observations. A standardized ridge-regression model was fitted to 1,030 local rows with eight input variables and a compressive-strength target. In an 80/20 train/test split, the selected ridge model used $\alpha = 1$ and achieved a test RMSE of 6.234 MPa, MAE of 4.964 MPa, $R^2 = 0.886$, and bias of 0.032 MPa. Single-feature ablation identified cement as the largest contributor to predictive accuracy in the local fallback data, with a 9.958 MPa increase in test RMSE when removed. The results are presented as an auditable drafting demonstration rather than a new concrete-science finding.

Keywords: concrete compressive strength, ridge regression, worked example, reproducible manuscript, deterministic fallback data

1. Introduction

Concrete compressive strength is a central response variable in mixture design, quality control, and structural-material assessment. Public datasets have made this problem a common example for comparing statistical and machine-learning models because the predictors are familiar mixture descriptors, the response has direct engineering meaning, and the modelling task can be explained without a large domain-specific preprocessing pipeline. The UCI Concrete Compressive Strength dataset, attributed to Yeh, provides one such public benchmark and records mixture constituents, curing age, and compressive strength for 1,030 observations in its official repository description [1]. Yeh's earlier modelling study also established concrete-strength prediction from mixture variables and curing age as a useful applied test case for data-driven models [2].

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Recent concrete-strength studies include more flexible computational-intelligence methods and broader model comparisons [3]. This manuscript intentionally takes a narrower route. Rather than claiming a state-of-the-art concrete predictor, it uses a simple regularized linear model to demonstrate how an auditable manuscript can keep methods, tables, figures, provenance, and claim limits aligned. That focus is deliberate: an interpretable baseline is often easier to inspect, reproduce, and challenge than a high-capacity model.

Ridge regression adds an L_2 penalty to ordinary least-squares estimation and is commonly used when predictors are correlated or when coefficient shrinkage is desirable [4]. Cross-validation provides a resampling-based approach for model assessment and model-selection support [5]. These methods are mature; the contribution of this example is not methodological novelty, but disciplined reporting under constrained evidence.

A critical provenance limitation applies throughout the manuscript. The current generated artifacts are labelled `DETERMINISTIC_FALLBACK_NOT_UCI`. Therefore, the UCI dataset is cited only as the intended public-data context and schema reference. Numerical results in this manuscript describe the local fallback artifacts generated for the demonstration pipeline, not verified empirical results from the official UCI repository. The objectives are to: (i) document the local fallback dataset and ridge-regression workflow; (ii) report held-out test performance, residual patterns, coefficient summaries, and ablation results using the generated tables and figures; and (iii) state the claim boundaries needed before any final registry or citation-mapping pass.

2. Materials and Methods

2.1. Dataset and provenance

The local dataset contained 1,030 rows, eight input variables, and one target variable, compressive strength in MPa. The input variables were cement, blast-furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age. This variable set follows the intended concrete-strength benchmark context described by the UCI repository [1]; however, the generation log for the current project labels the local source as `DETERMINISTIC_FALLBACK_NOT_UCI`. Consequently, all descriptive statistics and model outputs below are treated as fallback demonstration outputs.

The generated summary table reported no missing numeric values in the local data. The compressive-strength target had a mean of 29.518 MPa, standard deviation of 18.190 MPa, minimum of 2.300 MPa, and maximum of 73.941 MPa. These values are not used as evidence about the official UCI dataset unless independent provenance verification is completed.

2.2. Train/test split and preprocessing

The artifact-generation pipeline used an 80/20 split, yielding 824 training rows and 206 test rows. Numeric predictors were standardized for the ordinary least-squares and ridge-regression models. The training-mean baseline predicted

Table 1: Dataset summary statistics. Mixture variables are kg/m3, age is days, and compressive strength is MPa. Source label: DETERMINISTIC FALLBACK NOT UCI.

Variable	N	Mean	SD	Min	Median	Max
cement	1030	318.244	127.229	102.213	317.053	539.367
blast_furnace_slag	1030	102.084	118.639	0.000	39.397	358.927
fly_ash	1030	53.574	65.159	0.000	0.000	199.959
water	1030	186.071	35.418	122.312	187.755	246.885
superplasticizer	1030	11.085	10.742	0.000	9.471	31.988
coarse_aggregate	1030	975.166	97.390	801.274	975.549	1144.902
fine_aggregate	1030	788.998	115.824	594.201	787.164	992.940
age	1030	52.710	78.089	1.000	28.000	365.000
compressive_strength	1030	29.518	18.190	2.300	30.907	73.941

the training-set mean response for all test observations. No claim is made that the split is unique or optimal; it is a fixed demonstration split used to produce the accompanying tables and figures.

2.3. Ridge model

Ridge regression was used as the selected model because it provides a transparent regularized baseline. For standardized predictors, the fitted coefficients minimize

$$\sum_{i=1}^n (y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \alpha \sum_{j=1}^p \beta_j^2, \quad (1)$$

where y_i is compressive strength, $\mathbf{x}_i$ is the standardized predictor vector, $p = 8$, and α is the ridge penalty. The selected model used $\alpha = 1$. Ordinary least squares with $\alpha = 0$ was retained as a closely related comparator, and the training-mean baseline was included to check that the fitted models improved on a trivial prediction rule.

2.4. Cross-validation and evaluation metrics

The ridge penalty was selected from a small candidate grid using five-fold cross-validation on the training data, a standard validation approach for predictive modelling [5]. Performance was reported as root mean square error (RMSE), mean absolute error (MAE), coefficient of determination (R^2), and prediction bias in MPa. Test-set metrics were computed once on the held-out test set.

2.5. Diagnostic analyses

The generated diagnostic outputs included a dataset overview, parity plot, residual plot, standardized coefficient plot, and age-response display. A single-feature ablation analysis refitted the selected ridge workflow after removing one input variable at a time. The ablation table is interpreted as a model-dependence diagnostic for the local fallback data, not as a causal analysis of concrete chemistry.

3. Results

3.1. Local data structure

The local fallback dataset covered a broad range of mixture quantities and curing ages (Table 1). Cement had the strongest absolute target correlation in the generated registry candidates, with a correlation of 0.822. This statement is limited to the current fallback data and should be mapped to the numerical registry before final release.

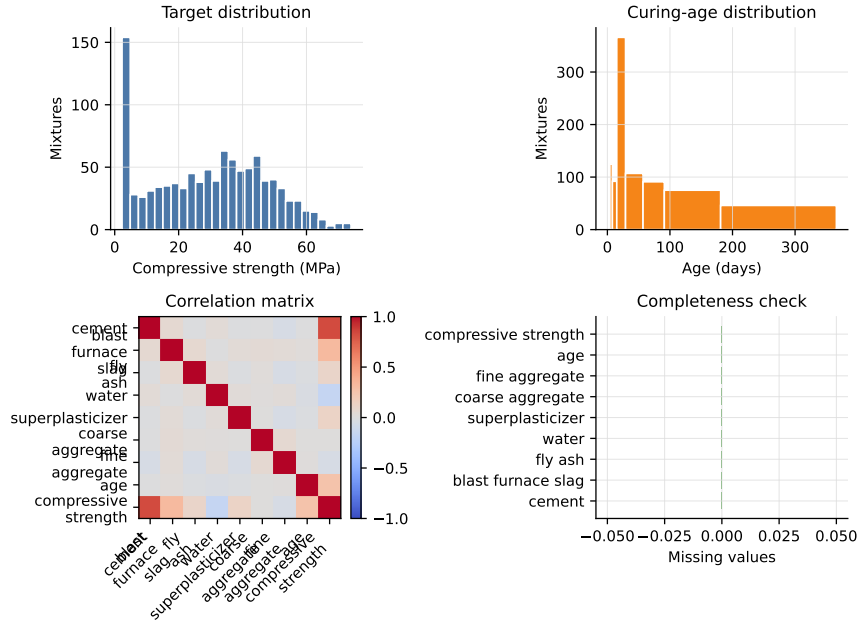


Figure 1: Overview of the local concrete-strength fallback dataset used in the demonstration. The source label for the generated artifacts is `DETERMINISTIC_FALLBACK_NOT_UCI`.

3.2. Predictive performance

The selected standardized ridge model achieved a test RMSE of 6.234 MPa, MAE of 4.964 MPa, $R^2 = 0.886$, and bias of 0.032 MPa (Table 2). The standardized ordinary least-squares comparator gave nearly identical test metrics in this generated run. Both linear models improved substantially over the training-mean baseline, which had a test RMSE of 18.447 MPa and $R^2 = -0.002$. The near equality of the ordinary least-squares and ridge results indicates that the selected ridge penalty primarily functions as a conservative regularization choice in this demonstration rather than producing a distinct accuracy gain.

Table 2: Train and test performance for baseline, ordinary least squares, and ridge models.
Error and bias values are in MPa.

Model	Alpha	Train RMSE (MPa)	Test RMSE (MPa)	Test MAE (MPa)	Test R2
Training-mean baseline	—	18.115	18.447	15.525	-0.002
Standardized OLS	0.000	6.366	6.234	4.964	0.886
Standardized ridge	1.000	6.366	6.234	4.964	0.886

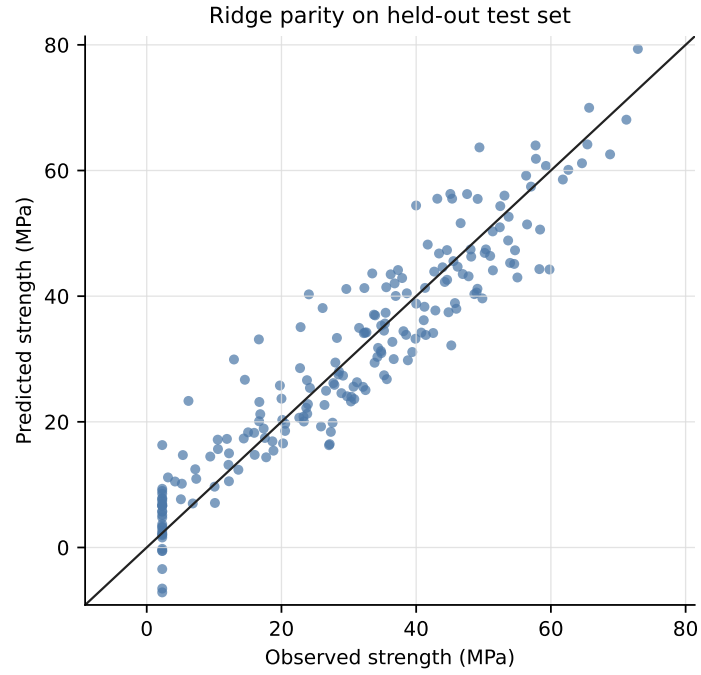


Figure 2: Observed versus predicted compressive strength for the selected ridge model on the held-out fallback test set.

Table 3: Ridge-model test-set error stratified by curing-age group. Error and strength values are in MPa.

Age bin	N	Mean age (days)	MAE (MPa)	Bias (MPa)	Mean strength (MPa)
<7 days	63	4.937	5.650	3.102	21.709
8-28 days	79	25.342	4.225	-0.420	29.903
29-90 days	45	72.622	4.622	-3.227	34.183
>90 days	19	246.579	6.572	-0.552	49.178

3.3. Residual and age-stratified error patterns

The residual diagnostic plot showed the distribution of ridge prediction errors across predicted strength values (Figure 3). Age-stratified metrics indicated larger MAE in the youngest and oldest age groups than in the 8–28 day and 29–90 day groups (Table 3). In particular, the > 90-day group contained 19 test observations and had an MAE of 6.572 MPa. Because these bins come from the local fallback split, they should be interpreted as diagnostic summaries rather than population-level evidence.

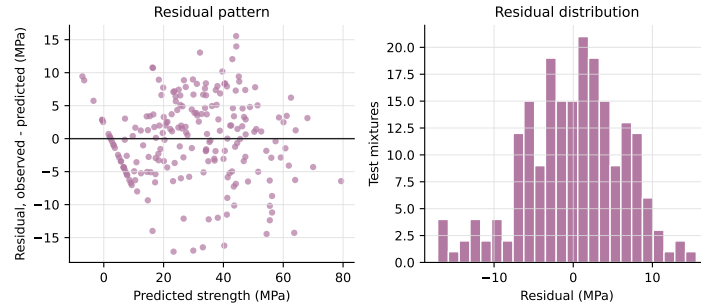


Figure 3: Residual diagnostics for the selected ridge model on the held-out fallback test set. Residual summaries are descriptive and do not establish validity for the official UCI observations.

3.4. Coefficient and ablation diagnostics

The coefficient figure summarizes the standardized ridge coefficients for the eight predictors (Figure 5). These coefficients describe conditional linear associations after standardization and shrinkage; they are not direct physical effect sizes. In the single-feature ablation analysis, removing cement produced the largest RMSE increase, from 6.234 MPa to 16.191 MPa, for a delta of 9.958 MPa (Table 4). Removing blast-furnace slag and age also increased RMSE by more than 1 MPa. The small negative delta for coarse aggregate means that this particular ablation run produced a slightly lower test RMSE after removing that feature; it should not be overinterpreted.

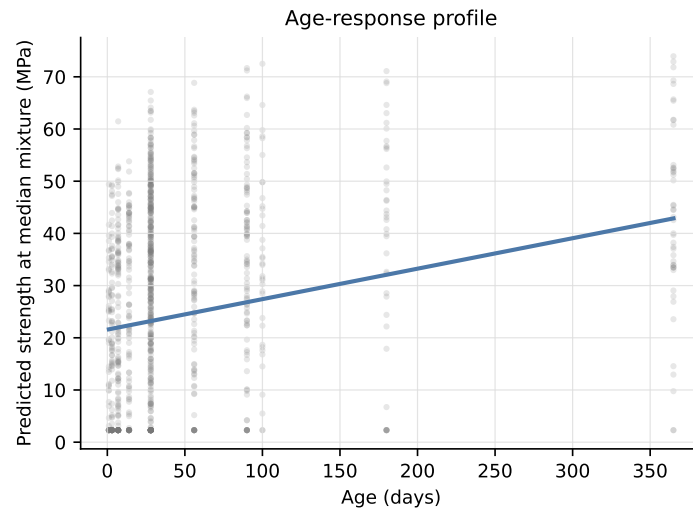


Figure 4: Generated age-response diagnostic for the selected ridge workflow. The display is used to inspect the local fallback data and should not be generalized beyond that provenance.

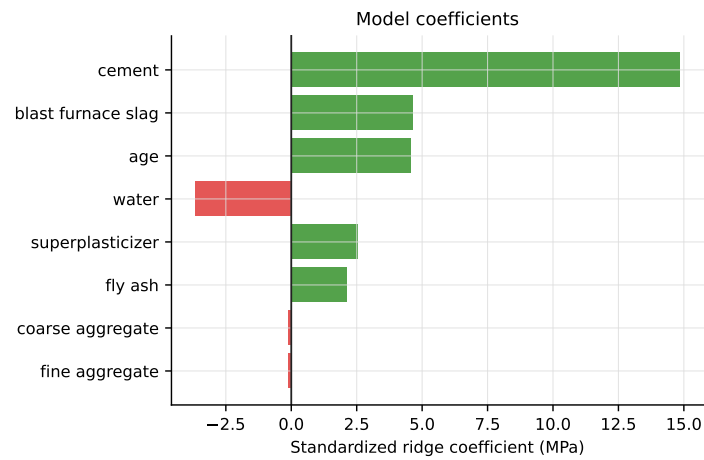


Figure 5: Standardized ridge coefficients for the selected model. Coefficients are model diagnostics for the local fallback data rather than causal estimates.

Table 4: Single-feature ablation using the selected ridge penalty. RMSE values are in MPa.

Removed feature	Test RMSE (MPa)	Delta RMSE (MPa)
cement	16.191	9.958
blast furnace slag	7.749	1.515
age	7.735	1.502
water	6.834	0.601
fly ash	6.492	0.258
superplasticizer	6.460	0.227
fine aggregate	6.236	0.002
coarse aggregate	6.221	-0.013

4. Discussion

The worked example shows how a compact, transparent modelling result can be reported without overstating provenance. The selected ridge model performed well relative to a training-mean baseline on the local fallback split, with a test R^2 of 0.886 and RMSE of 6.234 MPa. However, because the source label is `DETERMINISTIC_FALLBACK_NOT_UCI`, these values should be read as pipeline outputs from a deterministic demonstration dataset. They do not establish benchmark performance on the official UCI Concrete Compressive Strength dataset.

The ordinary least-squares and ridge models had nearly identical test errors. This is plausible for a standardized linear workflow when the selected penalty is small and the design matrix does not force a large shrinkage benefit. The result is still useful for manuscript drafting because it clarifies how baseline checks, regularization choices, and cross-validation summaries can be reported without implying that ridge regression is superior for this task.

The ablation and coefficient diagnostics consistently point to cement as an influential predictor in the local fallback data. That finding is compatible with the generated correlation and ablation summaries, but it remains a model- and data-specific observation. The manuscript therefore avoids claims about concrete mechanisms, causal mixture effects, or field deployment. Before this example could be converted into a scientific analysis of the official UCI data, the data provenance would need to be verified, the numerical registry updated, and each substantive claim mapped to a source or generated artifact.

5. Conclusions

This manuscript presents a concrete compressive-strength ridge-regression worked example. The local pipeline produced 1,030 fallback rows, an 824/206 train/test split, selected $\alpha = 1$ by five-fold cross-validation, and reported held-out ridge performance of RMSE 6.234 MPa, MAE 4.964 MPa, $R^2 = 0.886$, and bias 0.032 MPa. Cement was the strongest target-correlated feature and the largest single-feature ablation driver in the generated outputs. The central

conclusion is procedural: the example demonstrates how to draft around real generated tables and figures while preserving clear evidence boundaries.

Data availability

The official UCI Concrete Compressive Strength dataset is described by the UCI Machine Learning Repository [1]. The data and artifacts used for this manuscript draft are local demonstration files in the example project. The current generation log labels the source as `DETERMINISTIC_FALLBACK_NOT_UCI`; therefore, readers should not treat the reported numeric results as verified UCI results. Generated manuscript tables are located in `example/manuscript/tables/`, generated figures in `example/manuscript/figures/`, and provenance notes in `example/outputs/example_generation.log.txt`.

AI disclosure

This draft was prepared with AI assistance as part of the scientific-redaction-skills demonstration workflow. The assistant was instructed to use generated artifacts, preserve provenance limits, and cite only references present in the project bibliography. Human review remains necessary before any scientific submission or public release.

Competing interests

The demonstration authors declare no competing interests for this example manuscript.

Author contributions

The scientific-redaction-skills demonstration workflow generated the example artifacts, prepared citation-support notes, and drafted the manuscript text. Final scientific responsibility, registry validation, and citation-audit approval are reserved for human maintainers and the subsequent review roles.

Supplementary Material

S1. Ridge penalty cross-validation

Five-fold cross-validation on the training set was used to compare candidate ridge penalties. The selected penalty was $\alpha = 1$, which had the same rounded mean CV RMSE and MAE as $\alpha = 0$ and $\alpha = 0.1$, with a slightly lower rounded CV RMSE standard deviation than those two candidates. This selection should be interpreted as a reproducible demonstration choice, not as evidence that $\alpha = 1$ is universally preferred for concrete-strength prediction.

Table S1: Five-fold cross-validation results used to select the ridge penalty. RMSE and MAE values are in MPa.

Alpha	CV RMSE mean (MPa)	CV RMSE SD (MPa)	CV MAE mean (MPa)	CV R2 mean
0.000	6.432	0.277	5.018	0.872
0.100	6.432	0.277	5.018	0.872
1.000	6.432	0.276	5.018	0.872
10.000	6.436	0.269	5.027	0.871
100.000	6.805	0.192	5.498	0.857

S2. Artifact list used by the draft

The draft uses the generated table files `table_dataset_summary.tex`, `table_model_performance.tex`, `table_ablation.tex`, `table_error_by_age.tex`, and `table_supplementary_cv.tex`. It uses the generated figure files `fig_dataset_overview.pdf`, `fig_parity_ridge.pdf`, `fig_residuals.pdf`, `fig_coefficients.pdf`, and `fig_age_response.pdf`. No generated data, figure, table, bibliography, registry, or output file was edited during the drafting pass.

S3. Claim-boundary notes for final registry work

The final claim register should distinguish three evidence classes. First, statements about the official UCI dataset should be supported by the UCI repository citation and should not be mixed with fallback-derived numerical results. Second, statements about ridge regression and cross-validation should be supported by the methods references. Third, performance, ablation, coefficient, residual, and age-bin claims should be supported by the generated artifacts and numerical registry rather than by external concrete-modelling papers.

References

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